PRAJWAL GOTMARE

Senior AI Engineer · Generative AI, Agentic Systems & RAG
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PROFESSIONAL SUMMARY

Senior AI Engineer with **4.5 years of experience** across **Machine Learning, NLP, and Generative AI**, spanning industry roles and independent consulting. Builds secure **RAG and Agentic AI** applications using **Python, FastAPI, LangGraph, MCP, and AWS**. Integrates **hybrid retrieval, reranking, structured outputs, human-in-the-loop approvals, and tenant-aware access control**. Applies **Ragas** for LLM evaluation, **Langfuse** for tracing, and **adversarial testing** for safety to deliver **reliable, measurable AI solutions**.

TECHNICAL SKILLS

Programming & Backend: Python | SQL | FastAPI | Flask | REST API Development | JWT Authentication | Role-Based Access Control

Machine Learning & NLP: scikit-learn | XGBoost | LightGBM | PyTorch | TensorFlow | Hugging Face | Pandas | NumPy | spaCy | NLTK | Demand Forecasting | Time-Series Modeling | Model Benchmarking | Transformer-based Question Answering

Generative AI & Agent Frameworks: OpenAI APIs | Gemini | Anthropic Claude | LangChain | LangGraph | MCP | RAG Pipelines | Agentic AI | Tool Calling | Structured Outputs | Prompt Engineering | LLM Fine-tuning | LoRA | QLoRA

LLM Evaluation, Safety & Observability: Ragas | Langfuse | Prompt-Injection Testing | Tenant Isolation

Data Engineering & Retrieval: PySpark | Spark SQL | BigQuery | Airbyte | Qdrant | Milvus | Hybrid Retrieval | Reranking | Elasticsearch | OpenSearch | PostgreSQL | MySQL | MongoDB

Cloud & Deployment: Docker | AWS (Lambda, ECS/Fargate, S3, DynamoDB, SQS, API Gateway, Cognito, CloudWatch, SageMaker, CDK) | Azure AI

Document AI & Computer Vision: OCR | Document Information Extraction | OpenCV | EasyOCR

Additional Development Tools: React | TypeScript | Claude Code

PROFESSIONAL EXPERIENCE

Senior Data Scientist | AIThinkers *July 2025 – June 2026*

- **Virginia Tile — Ignite SEO AI Platform** — full-stack SEO content platform automating product descriptions and FAQs at scale.
  - Designed and implemented a production **RAG chatbot on Milvus** for the Virginia Tile platform, indexing **25,000+ products**.
  - Built with **React 18, TypeScript, FastAPI, MySQL, OpenAI GPT-4o-mini, and Gemini 2.5 Flash Lite**, generating product descriptions and FAQs from product metadata and technical PDFs.
  - Implemented content evaluation across **4 quality metrics** (readability, grammar, plagiarism indicators, AI-written likelihood) with a **consensus-based approval workflow** requiring 50%+ reviewer approval.
  - Built real-time **edit locking** (heartbeats, lock takeovers, automatic timeouts, SSE notifications), plus version tracking, feedback, likes, and publishing workflows.
  - Integrated **AWS Cognito** authentication across 3 roles and connected **OpenAI, Gemini, and Salsify** for generation, quality scoring, and publishing.
- **Decentralz — DCT Clinical Trials Data Pipeline** — AWS-native pipeline to ingest, enrich, index, and validate **ClinicalTrials.gov** data.
  - Designed a distributed architecture across **9 AWS services** (API Gateway, Lambda, ECS Fargate, SQS, S3, DynamoDB, OpenSearch, Valkey, CloudWatch), containerized with Docker, processing data in configurable batches (default **500 records**).
  - Built **4 processing stages** — condition enrichment, location normalization, trial indexing, and therapeutic-area aggregation — plus **3 parallel QA workflows** with a configurable **90% pass threshold** before marking runs ready for production.
  - Maintained run-specific and global **OpenSearch indices** for trials, locations, conditions, and therapeutic-area mappings, with **fine-grained access control**, enabling isolated, traceable validation of every execution.
  - Exposed **REST APIs** to start pipeline runs and check status; added operational safeguards (DynamoDB locks, SQS depth checks, retries, CloudWatch error aggregation) and generated QA evidence in **PDF and CSV** reports stored to S3.

AI Engineer | Klizer (DCKAP) *May 2024 – May 2025*

- **Marysville Marine Demand Forecasting Project** — AI-powered demand-forecasting and inventory-management system for a marine-parts e-commerce business.
  - Forecasted the top **1,000–10,000 SKUs** over **4–8 week horizons** for **~162,000 products** across **5 Marysville websites**, using **2011–2025** historical sales, inventory, and supplier data.
  - Benchmarked forecasting architectures including **TimesNet, TimeMixer, PatchTST, Amazon Chronos/Bolt, and TabPFN**, and engineered holiday/weekend features to improve demand prediction.
  - Designed a production data pipeline (**Epicor P21 + Magento → Airbyte → BigQuery → ETL → Forecasting**) and built a **FastAPI** backend exposing demand-forecast, inventory, and **OpenAI**-powered conversational-AI endpoints.
  - Implemented inventory analytics — stock availability, days-of-stock, turnover, safety stock, and reorder-point calculations — to support supply-chain decision-making.

Senior Software Engineer | Techno Exponent *May 2023 – April 2024*

- **Sanskrit Teaching Platform (Aanantya)** — conversational bot making Sanskrit learning accessible to a diverse, global audience.
  - Built with **Python** and the **OpenAI API** to power an interactive Sanskrit-learning chatbot, enhanced to converse in **15+ global languages**, expanding user acquisition.

- **Caller Support Bot** — AI-powered customer support bot for automated appointment booking.
  - Built with **Python, MySQL, OpenAI API, LangChain, and gTTS** for voice-based customer interactions, tailored for appointment booking across **hospitals, barbershops, and other service businesses**.

Independent AI Consultant | Freelance *November 2022 – April 2023*

- **AI-Powered Legal Decision-Support Platform | Mosu Tech** — ML-powered platform estimating cost, timeframe, and outcome across litigation, arbitration, and mediation.
  - Addressed uncertainty around **cost, resolution time, success probability, and lawyer selection** in legal disputes across **3 dispute-resolution paths**.
  - Built prediction engines for cost, timeframe, and outcome/settlement probability, exposed via **Flask REST APIs**; created a lawyer-recommendation engine using expertise, location, practice area, language, and budget.
  - Processed **640+ lawyer records** and **100+ court-case records**; evaluated **Random Forest, LightGBM, SVR, KNN, Decision Trees, and neural networks**, achieving up to **0.49 R^2** during model benchmarking.

Data Scientist | Shyena Tech Yarns Pvt Ltd *August 2021 – October 2022*

- **Alexa Skill for the Shipping Industry** — voice-based Alexa skill that interpreted commands and delivered container freight rates.
  - Built with the **Alexa Skill Kit and NLP** to interpret voice commands and retrieve freight rates; increased user engagement by **25% within 3 months** of launch.
- **Intelligent Invoice Data Extraction System** — end-to-end Document AI and OCR pipeline for extracting structured fields from invoices, receipts, and multi-page PDFs.
  - Processed **362 PDFs** (incl. **60 multi-page PDFs**) and **118 labelled invoice images**, plus **10,800+ XML annotations** and **~12,000 document images**.
  - Applied **EasyOCR, OpenCV, TensorFlow Object Detection, and PyTorch** for region detection and text extraction across vendor names, dates, invoice numbers, and tabular data, exporting structured CSV, JSON, and Excel outputs.
- **Legal Contract Question-Answering System** — transformer-based NLP system for extracting key clauses from long legal contracts.
  - Fine-tuned **RoBERTa-based QA models** on the **CUAD dataset** (**510 contracts, 20,910 QA pairs**) using CUAD/SQuAD-compatible formats.
  - Built pipelines for tokenisation, answer-span prediction, and confidence scoring; tested across **41 legal-question categories**.
- **Grocery Recommendation & Multilingual Search API** — built a Flask recommendation and product-search API on **23,053 grocery transactions**, integrating **Elasticsearch, MongoDB, and SQL Server** with JWT authentication and multilingual search.
- **Adibuja PySpark Personalisation Engine** — built a **PySpark and Spark SQL** personalisation engine for an e-commerce platform, using window functions to generate category and subcategory recommendations from **30,896 records**.
- **Rasa Voice and HR Chatbot** — built a **Rasa** assistant for HR and loan workflows with custom actions and a Flask interface, extended into an Employee Engagement Bot that ran **25 surveys** over **100 days**.

Quantitative Technical Instructor & Subject Matter Expert | Freelance / Independent *June 2019 – August 2021*

- Delivered online instruction in Engineering Mechanics and Energy Engineering; leveraged university distinction (Energy Engineering Subject Topper) to mentor **50+ undergraduate students** in quantitative analysis and complex problem-solving.

SELECTED PROJECTS

Business Brain | Governed Operations Copilot *Personal Project*

Tech Stack: Python, FastAPI, LangGraph, MCP, Qdrant, Groq, Neon/PostgreSQL, Auth0, Langfuse, Vercel

- Built a personal operations copilot on **mostly synthetic data** that routes questions to fixed analytics, cited document search, or approval-gated drafts.
- Enforced **tenant and role checks** across fixed SQL and **5 read-only MCP tools**; loaded **9,392 synthetic rows** in **23 tenant-scoped tables**.
- Indexed **10 synthetic PDFs** into **49 clause-aware chunks**; combined **bge-small embeddings** and **Qdrant BM25** with **RRF fusion**, reranking, and citations.
- Ran a **50-case synthetic evaluation**; **96.9% Ragas Faithfulness** across **8 judged document answers**.
- Passed a **21-case synthetic adversarial suite** (tenant leakage, privilege escalation, prompt injection); external actions stay disabled.

EDUCATION & CERTIFICATIONS

Bachelor of Engineering — Savitribai Phule Pune University, Pune · CGPA 7.15/10 *June 2015 – August 2019*

Certifications: Tableau Desktop Professional | **Languages:** English, Hindi, Marathi (Full Professional Proficiency)